

P1: Aerogel Incorporated Cement-Based Mixtures

P2: Analysis of Prestressed Concrete Beam Ends

ENGINEERING CLINIC PROJECTS – FALL 2026

P1: Aerogel Incorporated Cement-Based Mixtures – Lomboy

Aerogels are the world's lightest solid materials



Water-repellent cementitious substrates



- a) Prepare and test mortar specimens**
- b) Develop protocols for incorporating aerogel in mortar cubes**
- c) Measure the effects of aerogel on the physical and mechanical properties of mortars**

P1: Aerogel Incorporated Cement-Based Mixtures – Lomboy

This project will explore strategies for integrating aerogels through both bulk incorporation into the concrete matrix and surface-rendering techniques.

Aerogels, with their ultralow thermal conductivity, high porosity, and intrinsically hydrophobic surfaces, can disrupt liquid water transport, reduce the degree of saturation within the pore network, and thereby blunt the mechanisms of freeze-thaw damage. By optimizing aerogel dosage and coupling it with appropriate agents and surface treatments, the project will engineer concrete that maintains mechanical performance while exhibiting superior durability.

If you have questions about the topic, please contact Dr. Lomboy. lomboy@rowan.edu

WHAT YOU WILL BE DOING –

Activities

- Review methods of making concrete hydrophobic and properties of aerogel
- Prepare materials for testing
- Mix mortar and make specimens
- Measure the physical and mechanical properties mortar specimens
- Make sure work areas are clean and orderly

Reports and Presentations

- Write a final report or paper and present at the end of the semester

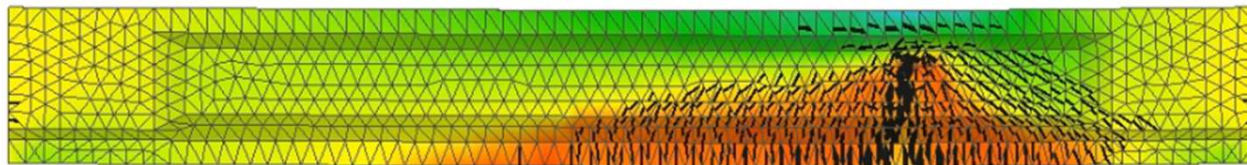
P2: Analysis of Prestressed Concrete Beam Ends— Lomboy



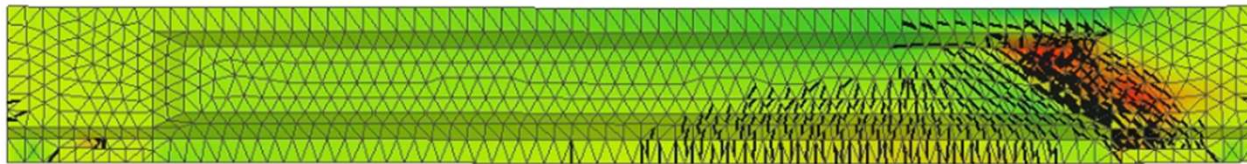
Deteriorated PSC beam end

Analyze the effects of end deterioration and repair on prestressed concrete beams by nonlinear finite element method

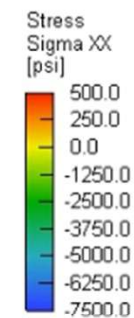
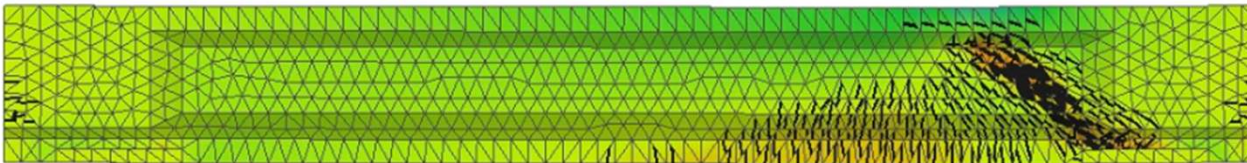
Undamaged



1ft Flange Damage



3ft Flange Damage



Beam failure modes simulated by FEM

P2: Analysis of Prestressed Concrete Beam Ends— Lomboy

One of the primary concerns with using precast prestressed concrete beams in bridges is cracking and spalling at the ends of the beams due to contamination from water and deicing chemicals, which exposes and corrodes the beams' reinforcement and prestressing strands. If allowed to progress, this deterioration will compromise the beams' capacity, which can affect the integrity of the entire bridge and raise safety and durability concerns. This project will analyze the effects of beam-end damage using FEM. New Jersey beams will be modeled and analyzed considering the nonlinear effects of material failure and progressive damage during loading.

If you have questions about these requirements, please contact Dr. Lomboy:
lomboy@rowan.edu

WHAT YOU WILL BE DOING –

Activities

- Review beam end deterioration in PC beam ends in New Jersey
- Learn to use ATENA nonlinear finite element software
- Analyze the effects of damage and repair on PS concrete beams

Reports and Presentations

- Write a final report or paper and present at the end of the semester